SJF

Q1.Write a C program to simulate the SJF (Non preemptive).

OS LAB > C P1.c > main()

```c
#include<stdio.h>
int main()
{
    int n,i,j,time=0,smallest;
    int at[10],bt[10],ct[10],tat[10],wt[10];
    int completed[10]={0};
    printf("Enter number of processes: ");
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("P%d Arrival Time: ",i+1);
        scanf("%d",&at[i]);

        printf("P%d Burst Time: ",i+1);
        scanf("%d",&bt[i]);
    }
    int done=0;
    while(done<n)
    {
        smallest=-1;
        for(i=0;i<n;i++)
        {
            if(at[i]<=time && completed[i]==0)
            {
                if(smallest==-1 || bt[i]<bt[smallest])
                    smallest=i;
            }
        }
        if(smallest==-1)
        {
            time++;continue;
        }
        time+=bt[smallest];
        ct[smallest]=time;
        completed[smallest]=1;
        done++;
    }
    for(i=0;i<n;i++)
```

OS LAB > C P1.c > main()

```c
        }
        if(smallest==-1)
        {
            time++;continue;
        }
        time+=bt[smallest];
        ct[smallest]=time;
        completed[smallest]=1;
        done++;
    }
    for(i=0;i<n;i++)
    {
        tat[i]=ct[i]-at[i];
        wt[i]=tat[i]-bt[i];
    }
    printf("\nProcess\tAT\tBT\tCT\tTAT\tWT\n");
    for(i=0;i<n;i++)
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",i+1,at[i],bt[i],ct[i],tat[i],wt[i]);

}
```

```
cd "/Users/sharan/Documents/C/OS LAB/" && gcc P1.c -o P1 && "/Users/sharan/Documents/C/OS LAB/"P1
→  C cd "/Users/sharan/Documents/C/OS LAB/" && gcc P1.c -o P1 && "/Users/sharan/Documents/C/OS LAB/"P1
Enter number of processes: 5
P1 Arrival Time: 0
P1 Burst Time: 8
P2 Arrival Time: 1
P2 Burst Time: 1
P3 Arrival Time: 2
P3 Burst Time: 3
P4 Arrival Time: 3
P4 Burst Time: 2
P5 Arrival Time: 4
P5 Burst Time: 6

Process AT      BT      CT      TAT     WT
P1      0       8       8       8       0
P2      1       1       9       8       7
P3      2       3       14      12      9
P4      3       2       11      8       6
P5      4       6       20      16      10
→  OS LAB
```

SJF

Q2.Write a C program to simulate the SJF (Preemptive).

OS LAB > C P2.c > main()

```c
#include<stdio.h>
int main()
{
    int n,i,time=0,remain;
    int at[10],bt[10],rt[10];
    int ct[10],tat[10],wt[10];
    int smallest;
    printf("Enter number of processes: ");
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("P%d Arrival Time: ",i+1);
        scanf("%d",&at[i]);
        printf("P%d Burst Time: ",i+1);
        scanf("%d",&bt[i]);
        rt[i]=bt[i];
    }
    remain=n;
    while(remain!=0)
    {
        smallest=-1;
        for(i=0;i<n;i++)
        {
            if(at[i]<=time && rt[i]>0)
            {
                if(smallest==-1 || rt[i]<rt[smallest])
                    smallest=i;
            }
        }
        if(smallest==-1)
        {
            time++;
            continue;
        }
        rt[smallest]--;
        time++;
        if(rt[smallest]==0)
        {
```


```c
        if(smallest==-1)
        {
            time++;
            continue;
        }
        rt[smallest]--;
        time++;
        if(rt[smallest]==0)
        {
```


```c
            ct[smallest]=time;
            remain--;
        }
    }
    for(i=0;i<n;i++)
    {
        tat[i]=ct[i]-at[i];
        wt[i]=tat[i]-bt[i];
    }
    printf("\nProcess\tAT\tBT\tCT\tTAT\tWT\n");
    for(i=0;i<n;i++)
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",i+1,at[i],bt[i],ct[i],tat[i],wt[i]);

}
```

```
→  C cd "/Users/sharan/Documents/C/OS LAB/" && gcc p2.c -o p2 && "/Users/sharan/Documents/C/OS LAB/"p2
Enter number of processes: 5
P1 Arrival Time: 0
P1 Burst Time: 8
P2 Arrival Time: 1
P2 Burst Time: 1
P3 Arrival Time: 2
P3 Burst Time: 3
P4 Arrival Time: 3
P4 Burst Time: 2
P5 Arrival Time: 4
P5 Burst Time: 6

Process AT      BT      CT      TAT     WT
P1      0       8       20      20      12
P2      1       1       2       1       0
P3      2       3       5       3       0
P4      3       2       7       4       2
P5      4       6       13      9       3
→  OS LAB
```